

Introduction to Linux

7CSEF006W

Dr Ayman El Hajjar

Academic Year: 2025/26

Please read this

- ➡ **NOTE-** In this lab, you only need a Linux machine. You can use the **Kali Linux machine**.
- ➡ THIS IS NOT MEANT TO BE A LAB TO FOLLOW STEP BY STEP. THIS IS A TUTORIAL ABOUT THE MAIN LINUX COMMANDS THAT WE WILL BE USING IN THIS MODULE AT SOME POINT.

Introducing Linux

- ➡ Linux is more than an OS. It's an idea where everybody grows together and there's something for everybody.
- ➡ For this reason, there are many flavours and distributions for Linux.
- ➡ The two most commonly used architecture for those distributions are either RPM based or Debian based.
- ➡ **RPM Red Hat Linux** and **SUSE Linux** were the original major distributions that used the .rpm file format, which is today used in several package management systems.
- ➡ **Debian:** Debian is a distribution that emphasizes free software. It supports many hardware platforms. Debian and distributions based on it use the .deb package format[5] and the dpkg package manager and its frontends (such as apt-get or synaptic)
- ➡ The two popular Linux OSs, Kali and Ubuntu, are Debian. In this lab we are using Kali Linux.

1 Learning the shell

➡ When we speak of the command line, we are really referring to the shell. The shell is a program that takes keyboard commands and passes them to the operating system to carry out.

➡ Almost all Linux distributions supply a shell program from the GNU Project called `bash`.

➡ When using a graphical user interface, we need another program called a terminal emulator to interact with the shell.

➡ It is called **terminal**

➡ In most cases, Linux errors are self explanatory. Read the error the terminal throws at you.

1.1 Simple Commands

➡ The `date` command displays the current time and date.

```
➤- date
```

➡ To see the current amount of free space on your disk drives, you use:

```
➤- df
```

➡ To display the amount of free memory you use:

```
➤- free
```

➡ This command remembers the last 500 commands by default in most Linux distributions.

```
➤- history
```

➡ If we press the up-arrow key we can see previous commands.

➡ We can end a terminal session by either closing the terminal emulator window. you can also the shell prompt by typing:

```
➤- exit
```

1.2 File system tree

➡ Linux, like other Unix-like systems, organizes files in a hierarchical directory structure.

➡ Files are arranged in a tree-like pattern of directories and subdirectories.

- ➡ The top-level directory is called the root, which contains files and further subdirectories.
- ➡ Unlike Windows, which has separate filesystem trees for each drive, Linux uses a single unified filesystem.
- ➡ Storage devices are mounted at specific points in this tree, managed by the system administrator.

All files in Linux are stored within a single filesystem hierarchy, structured like an inverted tree with the root at the top and directories and subdirectories branching below.

- ➡ `/usr` - Locally customised software. (e.g., logs, caches, databases, printer spools).
- ➡ `/bin` - Essential user commands.
- ➡ `/sbin` - System administration commands.
- ➡ `/etc` - Configuration files for the system.
- ➡ `/var` - Files that change dynamically
- ➡ `/home` - User home directories for personal data and settings.
- ➡ `/root` - Home directory for the administrative superuser.
- ➡ `/tmp` - Temporary storage; files older than 10 days are auto-deleted.

2 Getting Help

- ➡ To view a command's manual page:

```
>- man command
```

- ➡ To get a quick summary of available options:

```
>- command -help
```

2.1 Commands to navigate file systems

Commands options

- ➡ Most commands use options consisting of a single character preceded by a dash, such as `-l`.
- ➡ But many commands, including those from the GNU Project, also support long options, consisting of a word preceded by two dashes.
- ➡ If you want to know what options you can use for specific commands, you can read the manual for each command by using the manual command **man**.
- ➡ For example, to see the manual for the `ls` command including all options for `ls`.

```
>- man ls
```

- ➡ To display the current working directory:

>- pwd

➞ To list the files and directories in the current working directory:

>- ls

➞ We can check the manual of the **ls** command to see what options we can use with it. Different options will allow us to see the output in a different format.

↳ **ls -a** or **ls --all** List all files, even those with names that begin with a period, which are normally not listed (i.e., hidden).

↳ **-d --directory** Changes the working directory to the previous working directory.

↳ **-F --classify** This option will append an indicator character to the end of each listed name (for example, a forward slash if the name is a directory).

↳ **-h --human-readable** In long format listings, display file sizes in human-readable format rather than in bytes.

↳ **-l** display its results in long format. This format contains a great deal of useful information.

➞ **cd**: To change your working directory (where we are standing in our tree-shaped maze) use the **cd** command. Type **cd** followed by the pathname of the desired working directory.

➞ To change the working directory to the **/bin** directory.

>- cd /bin

➞ To change the working directory to your home directory.

>- cd

➞ To change the working directory to the previous working directory.

>- cd -

➞ To move to the parent of the current working directory.

>- cd ..

➞ To change the working directory to the current user's home directory.

>- cd ~

➞ Use the **file** command to determine a file's type.

>- file /bin/ls

➞ The **less** command is a pager to view text files. Once started, it allows scrolling forward and backward. For example, to examine the file that defines system user accounts:

```
➤- less /etc/passwd
```

3 Checking Your Login with whoami

➡ If you have forgotten whether you are logged in as root or another user, use **whoami**:

```
➤- whoami
```

4 Manipulating Files and Directories

➡ You can use **mkdir** to create a directory:

```
➤- mkdir lab1
```

↳ Use **man mkdir** to learn about available options.

➡ **cp**: Copy files or directories.

➡ To copy a file to another filename (creating a duplicate and keeping the original intact):

```
➤- cp report.txt report_backup.txt
```

↳ Use **man cp** to learn about available options.

➡ To move or rename a file or directory (without creating a duplicate):

```
➤- mv report.txt final_report.txt
```

↳ This basically moves report.txt and rename it to final_report.txt

↳ Use **man mv** to learn about available options.

➡ If you want to move a file into a folder:

```
➤- mv report.txt /home/kali/Desktop/final_report.txt
```

➡ Use **rm** to remove files and directories

```
➤- rm report.txt
```

↳ Use **man rm** to learn options (including how to delete non-empty directories).

5 Working with Text Files

➡ To open and edit a text file using the **nano** editor:

```
➤- nano filename
```

↳ Use **Ctrl+O** to save changes, **Ctrl+X** to exit.

6 Monitoring Processes

➡ To view running processes in a snapshot:

```
➤ ps aux
```

➡ To monitor processes in real time and view CPU/memory usage:

```
➤ top
```

6.1 Text manipulation

➡ To view the contents of a file, use the **cat** command. For example, to display the configuration file for **adduser**:

```
➤ cat /etc/adduser.conf
```

➡ To view only the first few lines of a file, use the **head** command with the file name:

```
➤ head /etc/adduser.conf
```

➡ To view a specific number of lines from the top of the file, for example the first 20 lines:

```
➤ head -20 /etc/adduser.conf
```

➡ To view only the last few lines of a file, use the **tail** command:

```
➤ tail /etc/adduser.conf
```

➡ To view a specific number of lines from the bottom of the file, for example the last 20 lines:

```
➤ tail -20 /etc/adduser.conf
```

➡ To display a file with line numbers alongside its content, use the **nl** command:

```
➤ nl /etc/adduser.conf
```

➡ To search for specific content inside a file, use the **grep** command. For example, to display all lines containing the word **groups**:

```
➤ cat /etc/adduser.conf | grep groups
```

7 Permission

- ➡ To understand Linux permissions, remember that it is a multitasking and multiuser operating system. Multiple people can use the same computer at the same time (for example, via **ssh**). The X Window System even supports remote graphical output.
- ➡ To manage multiple users securely, Linux isolates each user's processes and files from others. This multiuser capability has been part of Linux's design since the early days of Unix, when large central systems served many users through terminals.
- ➡ To check file and directory permissions, use the **ls** command. Its output shows a file mode string that defines read, write, and execute access for three groups: the owner, the group, and others.

① ② ③ ④
-rwxrw-r--

- ① File type
- ② Owner permissions
- ③ Group permissions
- ④ World permissions

- ➡ To understand the permissions format, note that the nine characters after the file type represent **read** (r), **write** (w), and **execute** (x) permissions for the owner, group, and everyone else. Permissions are modified using the **chmod** command.

Attribute	Files	Directories
r	Allows a file to be opened and read.	Allows a directory's contents to be listed if the execute attribute is also set.
w	Allows a file to be written to or truncated; however, this attribute does not allow files to be renamed or deleted. The ability to delete or rename files is determined by directory attributes.	Allows files within a directory to be created, deleted, and renamed if the execute attribute is also set.
x	Allows a file to be treated as a program and executed. Program files written in scripting languages must also be set as readable to be executed.	Allows a directory to be entered; e.g., <i>cd directory</i> .

- ➡ To change the permissions of a file or directory, use the **chmod** command. Only the file owner or the superuser can change permissions. **chmod** supports two formats: octal numbers and symbolic representation. In this lab, we focus on octal numbers.
- ➡ To understand directory permissions, note that they work differently than file permissions. Directories control who can list, create, or delete files within them.

Permission	Action	chmod option
read	(view contents, i.e. ls command)	r or 4
write	(create or remove files from dir)	w or 2
execute	(cd into directory)	x or 1

- ➡ To set permissions on a file called **filename** so that the **owner** has execute, read, and write access, the **group** has read and write access, and **others** have read access:

```
❯ chmod 764 filename
```

↳ where:

↳ 7 = read + write + execute (4+2+1)

↳ 6 = read + write (4+2)

↳ 4 = read only (4)

Superuser Commands

- ➡ To perform certain administrative tasks, such as installing software, editing system configuration files, or accessing restricted devices, you need superuser privileges.
- ➡ To run a command with elevated permissions, use the **sudo** command. You will be prompted for your password, but nothing will appear as you type it — this is intentional to hide the password length.
- ➡ To copy **run.sh** into the **/bin** directory with root privileges:

```
❯ sudo cp run.sh /bin
```

su vs sudo

Do not use the **su** command for routine tasks. It grants a persistent root shell without further password prompts, increasing the risk of accidental system damage. Instead, use **sudo** (**super user do**) to run only the commands that require elevated permissions.

8 System Information

- ➞ To display detailed system and kernel information:

```
➤- uname -a
```

- ➞ To check overall disk usage in human-readable format:

```
➤- df -h
```

Networking

IP Addresses

It is important to know the IP address of each virtual machine (VM) you are working on.

- ➞ To list all network interfaces and their assigned IP addresses:

```
➤- ip a
```

- ➞ The output may include physical interfaces, wireless adapters, and virtual interfaces. On the Kali machine used in this module, several interfaces may appear due to Docker containers. These can be ignored for now.

- ➞ To bring down a specific interface, such as `eth0`:

```
➤- ifconfig eth0 down
```

- ➞ To bring the same interface back up:

```
➤- ifconfig eth0 up
```

- ➞ To request a dynamic IP address from the DHCP server for `eth0`:

```
➤- dhclient eth0
```

- ➞ To check whether your machine has a wireless interface:

```
➤- iwconfig
```

↳ Many university lab machines do not have wireless interfaces (except those in CLG.43 and CLG.45). If none are present, you will see `No Wireless Extensions`.

- ➞ Ncat is a versatile command-line tool used for testing network connections, transferring data, and troubleshooting. It supports IPv4, IPv6, TCP, UDP, and SSL protocols.

- ➞ To install Ncat:

```
➤- sudo apt-get install ncat
```

➞ To listen for incoming TCP connections on port 8080:

```
➤ ncat -l 8080
```

➞ To connect to a remote system at 192.168.144.1 on port 80:

```
➤ ncat 192.168.144.1 80
```

➞ To listen for UDP connections on port 1234:

```
➤ ncat -l -u 1234
```

➞ To set a timeout of 10 seconds when connecting to 192.168.144.1 on port 8080:

```
➤ ncat -w 10 192.168.144.1 8080
```

➞ For more examples and advanced usage, see the [Quick Reference Guide](#).

9 Adding and Removing Software

➞ To update the repository list and ensure you have the latest software links:

```
➤ sudo apt-get update
```

➞ To edit the repository list manually, the APT source file is located at `/etc/apt/sources.list`.

➞ To open the file in a graphical editor:

```
➤ sudo mousepad /etc/apt/sources.list
```

➞ To open the file in a terminal-based editor:

```
➤ sudo nano /etc/apt/sources.list
```

➞ Linux uses the **Advanced Packaging Tool (APT)** to manage software installation and removal. You can use either **apt** or **apt-get**, but **apt-get** provides more advanced options.

➞ To install a software package (replace `pgplot5` with the desired package name):

```
➤ sudo apt-get install pgplot5
```

➞ To remove a software package while keeping its configuration files:

```
➤ sudo apt-get remove pgplot5
```

➞ To remove a software package along with its configuration files:

```
➤ sudo apt-get purge pgplot5
```

➡ To install software from GitHub, first install Git:

```
➤- sudo apt-get install git
```

➡ To clone a repository from GitHub (for example, the SIGIT tool):

```
➤- sudo git clone https://github.com/termuxhackers-id/SIGIT.git
```